

WORK SHEET - 12

ELLIPSE & HYPERBOLA

Q.1 $e > \frac{2}{\sqrt{3}}$

Sol. Let the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and any double ordinate PQ be $(a \sec \theta, b \tan \theta)$, $(a \sec \theta, -b \tan \theta)$.

O is the centre $(0, 0)$.
 ΔOPQ is equilateral. So

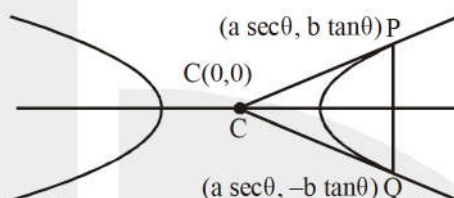
$$\tan 30^\circ = \frac{b \tan \theta}{a \sec \theta}$$

$$\Rightarrow 3 \frac{b^2}{a^2} = \operatorname{cosec}^2 \theta$$

$$\Rightarrow 3(e^2 - 1) = \operatorname{cosec}^2 \theta$$

$$\text{Now } \operatorname{cosec}^2 \theta \geq 1$$

$$\Rightarrow 3(e^2 - 1) \geq 1 \Rightarrow e^2 \geq \frac{4}{3} \Rightarrow e > \frac{2}{\sqrt{3}}. \text{ Ans.}$$



Q.2 $x^2 + y^2 = 5$

Sol. As, ellipse and hyperbola intersect at right angles, so they are confocal (i.e. same focii)

For ellipse, $\frac{x^2}{9} + \frac{y^2}{4} = 1$

$$e^2 = 1 - \frac{4}{9} = \frac{5}{9} \Rightarrow e = \frac{\sqrt{5}}{3}$$

$$\therefore \text{Focii of ellipse} = (\pm \sqrt{5}, 0)$$

For hyperbola, $\frac{x^2}{4/\alpha^2} - \frac{y^2}{4} = 1$

$$e_h^2 = 1 + \frac{4}{\left(\frac{4}{\alpha^2}\right)} = 1 + \alpha^2 \Rightarrow e_h = \sqrt{1 + \alpha^2}$$

$$\therefore \text{Focii of hyperbola} = \left(\pm \frac{2}{\alpha} \sqrt{1 + \alpha^2}, 0 \right)$$

Now, according to question,

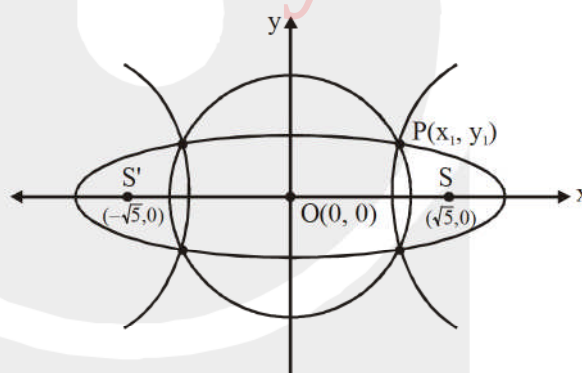
$$\frac{2}{\alpha} \sqrt{1 + \alpha^2} = \sqrt{5} \Rightarrow \frac{4}{\alpha^2} (1 + \alpha^2) = 5 \Rightarrow \frac{4}{\alpha^2} + 4 = 5 \Rightarrow \alpha^2 = 4$$

$$\therefore a = 2$$

So, equation of hyperbola becomes

$$4x^2 - y^2 = 4 \quad \dots\dots(1)$$

and equation of given ellipse is



$$4x^2 + 9y^2 = 36 \quad \dots\dots(2)$$

Let point of intersection in the first quadrant be $P(x_1, y_1)$. As P lies on both the curves, so

$$4x_1^2 + 9y_1^2 = 36 \quad \text{and} \quad 4x_1^2 - y_1^2 = 4$$

$\therefore$ On adding, we get

$$8(x_1^2 + y_1^2) = 40 \quad \Rightarrow \quad x_1^2 + y_1^2 = 5$$

Hence, the required equation of circle is $x^2 + y^2 = 5$, whose centre is at $(0, 0)$ and radius = $\sqrt{5}$

Q.3 $\frac{\sqrt{5}-1}{2}$

Sol. For $n=2$, here ellipse E_2 touches ellipse E_1 at the ends of major axis of ellipse E_1 and the foci of ellipse E_2 coincide with the ends of minor axis of ellipse E_1 . And so on.

(eccentricity = e_1) Let $E_1: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$

(eccentricity = e_2) $E_2: \frac{x^2}{A^2} + \frac{y^2}{B^2} = 1 \quad (B > A)$

And so on.

(Think?)

As ellipse E_2 meets x -axis at $(\pm A, 0)$. So according to question

$$A = a \quad \dots\dots(1)$$

Also, $e_2^2 = 1 - \frac{A^2}{B^2} \Rightarrow e_2 = \frac{\sqrt{B^2 - A^2}}{B}$

So, foci of $E_2 = (0, \pm \sqrt{B^2 - A^2})$

But, according to question, we are given

$$\sqrt{B^2 - A^2} = b \Rightarrow B^2 - A^2 = b^2 \quad (\text{squaring on both sides})$$

$$\therefore B^2 = a^2 + b^2 \quad \dots\dots(2) \quad (\text{As } A = a \text{ using (i)})$$

So, equation of ellipse E_2 is

$$\frac{x^2}{a^2} + \frac{y^2}{a^2 + b^2} = 1$$

As, eccentricity of ellipses is independent of n and e

$$\therefore e_1^2 = e_2^2 \Rightarrow \left(1 - \frac{b^2}{a^2}\right) = \left(1 - \frac{A^2}{B^2}\right) \Rightarrow \left(1 - \frac{b^2}{a^2}\right) = \left(1 - \frac{a^2}{a^2 + b^2}\right)$$

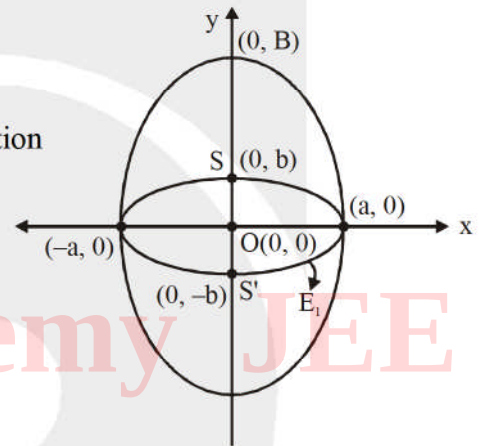
$$\Rightarrow \frac{b^2}{a^2} \cdot \left(1 + \frac{b^2}{a^2}\right) = 1 \quad \dots\dots(3)$$

As, $e_1^2 = 1 - \frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = (1 - e_1^2)$

$\therefore$ Equation (3) becomes

$$(1 - e_1^2)(2 - e_1^2) - 1 = 0 \Rightarrow e_1^4 - 3e_1^2 + 1 = 0$$

$$\therefore e_1^2 = \frac{3 \pm \sqrt{5}}{2}$$



But, $e_1^2 = \frac{3+\sqrt{5}}{2}$ (rejected)

$$\therefore e_1^2 = \frac{3-\sqrt{5}}{2} \Rightarrow e_1 = \sqrt{\frac{3-\sqrt{5}}{2}}$$

$$\Rightarrow e_1 = \sqrt{\frac{(3-\sqrt{5}) \times 2}{2 \times 2}} = \sqrt{\frac{6-2\sqrt{5}}{4}} = \sqrt{\frac{(\sqrt{5}-1)^2}{4}} = \frac{\sqrt{5}-1}{2} = 2 \sin 18^\circ \text{ Ans.}$$

Alter $1 - e_1^2 = \frac{b_1^2}{a_1^2}$

$$b_1 = b_2 e_2$$

$$1 - e_2^2 = \frac{a_1^2}{b_2^2}$$

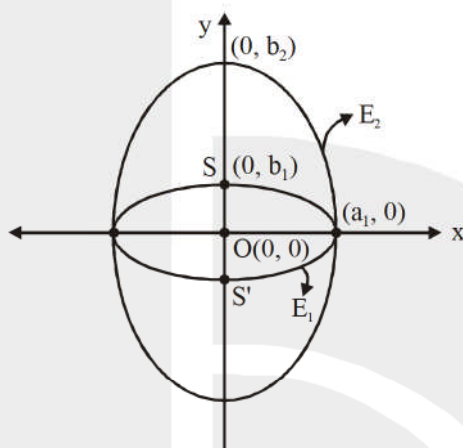
$$\Rightarrow 1 - e_2^2 = \frac{a_1^2}{b_2^2} e_2^2$$

$$\Rightarrow 1 - e^2 = \frac{1}{(1 - e^2)} e^2 \quad (e_1 = e_2 = e)$$

$$\Rightarrow (1 - e^2) = e$$

$$\Rightarrow e^2 + e - 1 = 0$$

$$\Rightarrow e = \frac{\sqrt{5}-1}{2}$$



Q.4 $\frac{x^2}{9} + \frac{y^2}{4} = \frac{1}{2}$

Sol. Let $P(a \cos \theta, b \sin \theta)$ and $Q(-a \sin \theta, b \cos \theta)$

Let $M(h, k)$ be the midpoint of PQ .

$$\text{So } \left(\frac{2h}{a}\right) = (\cos \theta - \sin \theta) \quad \dots\dots(1)$$

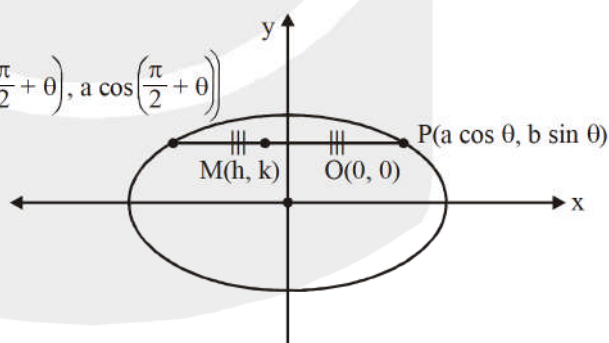
$$\text{Also } \left(\frac{2k}{b}\right) = (\sin \theta + \cos \theta) \quad \dots\dots(2)$$

$\therefore$ On squaring and adding, we get

$$\frac{4h^2}{a^2} + \frac{4k^2}{b^2} = 2$$

So, locus of $M(h, k)$ is

$$\frac{x^2}{3^2} + \frac{y^2}{2^2} = \frac{1}{2} \text{ Ans.}$$



Q.5 $\frac{(3x+y-8)^2}{1} - \frac{(x-3y+14)^2}{4} = 20$

Sol. $2ae = SS'$

$\therefore ae = \sqrt{10}$ (1)

and product of distance from focus on y-axis gives b^2

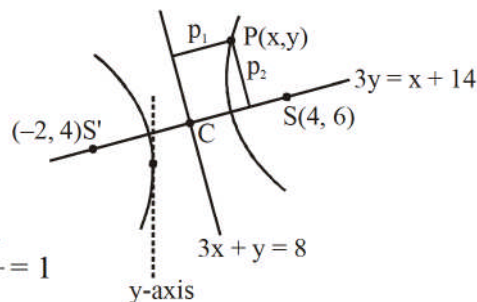
$\therefore b^2 = 8$ (2)

$\therefore a^2 = 2$, [$\because b^2 + a^2 = a^2 e^2$]

$\therefore$ Equation of hyperbola is

$$\frac{p_1^2}{a^2} - \frac{p_2^2}{b^2} = 1 \Rightarrow \frac{(3x+y-8)^2}{(\sqrt{10})^2 \cdot 2} - \frac{(x-3y+14)^2}{(\sqrt{10})^2 \cdot 8} = 1$$

i.e. $\frac{(3x+y-8)^2}{20} - \frac{(x-3y+14)^2}{80} = 1$ Ans.



Q.6 $\frac{1}{\sqrt{2}}$

Sol. Given equation of ellipse as $x^2 + 4y^2 = 4$ (1)

and equation of circle as $x^2 + y^2 = 2$ (2)

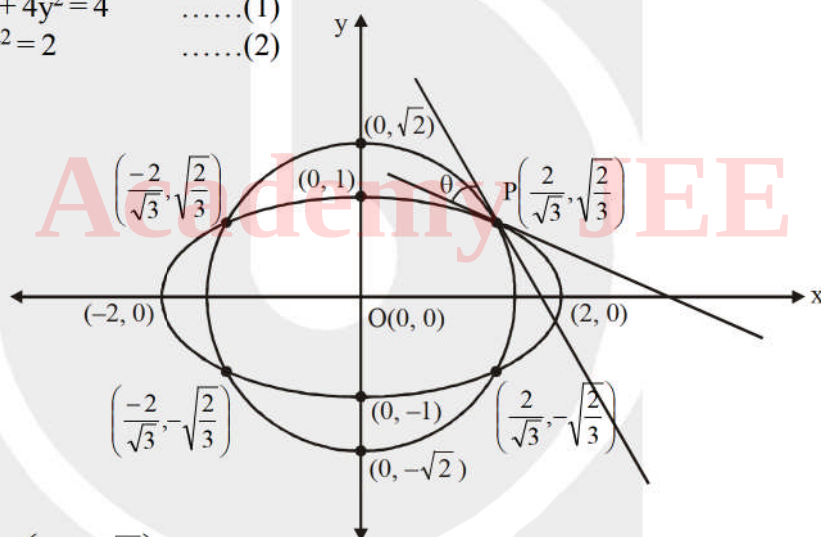
$\therefore$ Equation (1) – Equation (2)

$\Rightarrow 3y^2 = 2 \Rightarrow y = \pm \sqrt{\frac{2}{3}}$

Also, $x^2 = 2 - \frac{2}{3}$

$\Rightarrow x^2 = \frac{4}{3}$

$\Rightarrow x = \pm \frac{2}{\sqrt{3}}$



Let m_1 = slope of tangent at $P\left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}}\right)$ on ellipse

$\Rightarrow m_1 = \frac{-1}{2\sqrt{2}}$

Also M_2 = slope of tangent at $P\left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}}\right)$ on circle

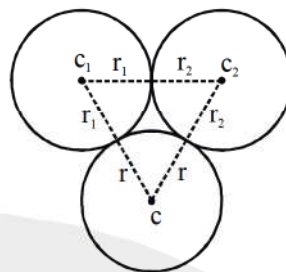
$\Rightarrow m_2 = -\sqrt{2}$

Let θ be the acute angle between tangents at $P\left(\frac{2}{\sqrt{3}}, \sqrt{\frac{2}{3}}\right)$ on

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{-1}{2\sqrt{2}} + \sqrt{2}}{1 + \left(\frac{1}{2\sqrt{2}}\right) \cdot \sqrt{2}} \right| = \left| \frac{\frac{3}{2\sqrt{2}}}{\frac{3}{2}} \right| = \frac{1}{\sqrt{2}} \quad \text{Ans.}$$

Q.7 2

Sol. $cc_1 = r + r_1$
 $cc_2 = r + r_2$
 $cc_1 - cc_2 = r_1 - r_2$
 $\Rightarrow$ Locus of c is hyperbola with foci c_1 and c_2
 $\Rightarrow 2ae = c_1 c_2 = r_1 + r_2$ and $r_1 - r_2 = 2a$
 $\Rightarrow \frac{r_1 + r_2}{r_1 - r_2} = \sqrt{2}$
 $\Rightarrow e^2 = 2$



Q.8 10

Sol. Given, $4ax + 3by = 24$ (1)
 is normal to the ellipse at $(a \cos \theta, b \sin \theta)$. So equation of normal to ellipse at $(a \cos \theta, b \sin \theta)$ is
 $a x \sec \theta - by \operatorname{cosec} \theta = (a^2 - b^2)$ (2)
 As (1) and (2) are identical, so

$$\frac{\sec \theta}{4} = \frac{-\operatorname{cosec} \theta}{3} = \frac{(a^2 - b^2)}{24}$$

$$\therefore \sec \theta = \frac{(a^2 - b^2)}{6} \Rightarrow \cos \theta = \left(\frac{6}{a^2 - b^2} \right)$$

$$\text{Also, } \operatorname{cosec} \theta = \frac{-(a^2 - b^2)}{8} \Rightarrow \sin \theta = \left(\frac{-8}{a^2 - b^2} \right)$$

Now, using $\sin^2 \theta + \cos^2 \theta = 1$

$$\Rightarrow \frac{64}{(a^2 - b^2)^2} + \frac{36}{(a^2 - b^2)^2} = 1$$

$$\Rightarrow 100 = (a^2 - b^2)^2 \Rightarrow (a^2 - b^2) = 10 (\because a > b) \text{ Ans.}$$

Q.9 $x^2 + y^2 - 12x - 2y + 33 = 0$

Q.10 $\frac{6}{5}\sqrt{10}$

Sol. Given, $S(2, 1)$, $S'(10, 1)$ are focii of hyperbola. Also $x + y - 9 = 0$ is tangent to this hyperbola
 As, centre of hyperbola = midpoint of SS'
 $\Rightarrow$ Centre of hyperbola = $(6, 1)$
 Let the equation of hyperbola be

$$\frac{(x-6)^2}{a^2} - \frac{(y-1)^2}{b^2} = 1, \quad \text{where}$$

$$e_h^2 = 1 + \frac{b^2}{a^2} \quad \text{.....(1)}$$

Now, using property

$$p_1 p_2 = b^2$$

$$\Rightarrow \frac{|2+1-9|}{\sqrt{2}} \times \frac{|10+1-9|}{\sqrt{2}} = b^2 \Rightarrow \left(\frac{6}{\sqrt{2}}\right)\left(\frac{2}{\sqrt{2}}\right) = b^2 \Rightarrow b^2 = 6$$

Also, distance between foci

$$\Rightarrow 2a e_h = 8 \Rightarrow a \cdot e_h = 4$$

$\therefore$ From (1), we get

$$a^2 \cdot e_h^2 = a^2 + b^2 \Rightarrow 16 = a^2 + 6 \Rightarrow a^2 = 10$$

So, the equation of hyperbola becomes

$$\frac{(x-6)^2}{10} - \frac{(y-1)^2}{6} = 1$$

(i) The equation of director circle of hyperbola is

$$(x-6)^2 + (y-1)^2 = a^2 - b^2$$

$$\Rightarrow (x-6)^2 + (y-1)^2 = 10 - 6$$

$$\Rightarrow x^2 + y^2 - 12x - 2y + 33 = 0 \quad \text{Ans.}$$

(ii) Length of latus rectum of hyperbola

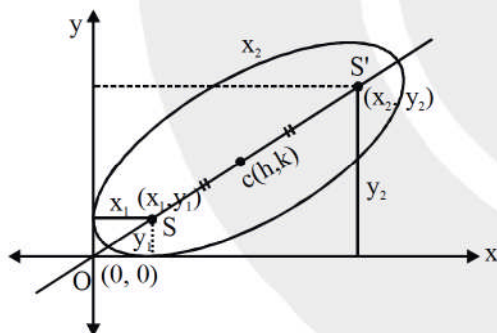
$$= \left(\frac{2b^2}{a}\right) = \frac{2 \times 6}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}} = \frac{6\sqrt{10}}{5} \quad \text{Ans.}$$

Q.11 $x^2 + y^2 = 5$

Q.12 $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 16$

Sol.

(i)



We know that $SS' = 2ae \Rightarrow (x_1 - x_2)^2 + (y_1 - y_2)^2 = 4(2)^2 \left(\frac{\sqrt{3}}{2}\right)^2$

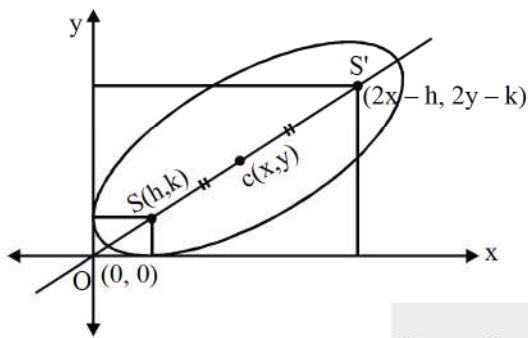
$$\Rightarrow (x_1 + x_2)^2 + (y_1 + y_2)^2 - 4(x_1 x_2 + y_1 y_2) = 12 \Rightarrow (2h)^2 + (2k)^2 - 4(1 + 1) = 12$$

(As, $x_1 \cdot x_2$ and $y_1 y_2$ are perpendicular distances of the foci from their tangents $= b^2 = 1^2 = 1$)

$$\Rightarrow h^2 + k^2 = 5$$

$\therefore$ The locus of centre $c(h, k)$ of ellipse is $x^2 + y^2 = 5$

(ii)



Now, $(2x-h)h = 1 \Rightarrow x = \left(\frac{1+h^2}{2h}\right)$; $(2y-k)k = 1 \Rightarrow y = \left(\frac{1+k^2}{2k}\right)$

Also, $x^2 + y^2 = 5 \Rightarrow \frac{(1+h^2)^2}{4h^2} + \frac{(1+k^2)^2}{4k^2} = 5$

$\therefore$ The locus of focus $S(h, k)$ is

$$x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 16.$$

Q.13 $\sqrt{\frac{7}{3}}$

Q.14 $2\sqrt{3}$

Sol.

(i) $B^2 = A^2(e_2^2 - 1)$ (i)

$b^2 = a^2(1 - e_1^2)$ (ii)

foci of ellipse $(\pm ae, 0)$

foci of hyperbola $(\pm Ae_2, 0)$

According to question $ae_1 = A$ (iii)

$Ae_2 = a$ (iv)

from (iii) and (iv)

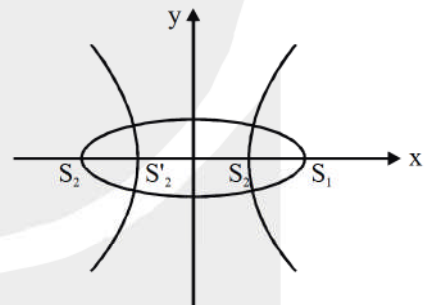
$$e_1 e_2 = 1 \Rightarrow e_1 = \frac{1}{2}$$

and using (i) and (ii) also, we get $a = \frac{2}{\sqrt{3}}$; $B^2 = 1, b = 1$

$\therefore$ Director circle of ellipse is $x^2 + y^2 = a^2 + b^2$

$$\Rightarrow \text{Radius of director circle} = \sqrt{a^2 + b^2} = \sqrt{\frac{7}{3}}$$

(ii) l (latus rectum of hyperbola) $= \frac{2B^2}{A} = 2\sqrt{3}$.



Q.15 (A) -7 (C) 8 (D) 10

Sol. The given ellipse is $\frac{x^2}{9} + \frac{y^2}{4} = 1$

Let the equation of line through $T(\alpha, 2)$ in parametric form is

$$\frac{x - \alpha}{\cos \theta} = \frac{y - 2}{\sin \theta} = r$$

Put $(x = \alpha - r \cos \theta, y = 2 - r \sin \theta)$ in the given (consider r as negative think)

Equation of ellipse, we get

$$\frac{(\alpha^2 + r^2 \cos^2 \theta - 2\alpha r \cos \theta)}{9} + \frac{(4 + r^2 \sin^2 \theta - 4r \sin \theta)}{4} = -1 = 0$$

$$\Rightarrow \left(\frac{\cos^2 \theta}{9} + \frac{\sin^2 \theta}{4} \right) r^2 + \left(\frac{-2\alpha}{9} \cos \theta - \sin \theta \right) r + \frac{\alpha^2}{9} = 0$$

$\swarrow \quad \searrow$
 TA TD

$$\therefore (TA)(TD) = \frac{\frac{\alpha^2}{9}}{\left(\frac{\cos^2 \theta}{9} + \frac{\sin^2 \theta}{4} \right)} = \frac{4\alpha^2}{4\cos^2 \theta + 9\sin^2 \theta} \dots\dots(1)$$

Again, put $(x = \alpha - r \cos \theta, y = 2 - r \sin \theta)$ in $xy = 0$ (line pair of coordinate axes)

$$\Rightarrow (\cos \theta \cdot \sin \theta) r^2 - (\alpha \sin \theta + 2 \cos \theta) r + 2\alpha = 0$$

$\swarrow \quad \searrow$
 (TB) (TC)

$$\therefore (TB)(TC) = \left(\frac{2\alpha}{\cos \theta \cdot \sin \theta} \right) \dots\dots(2)$$

Now, $(TA)(TD) = (TB)(TC)$

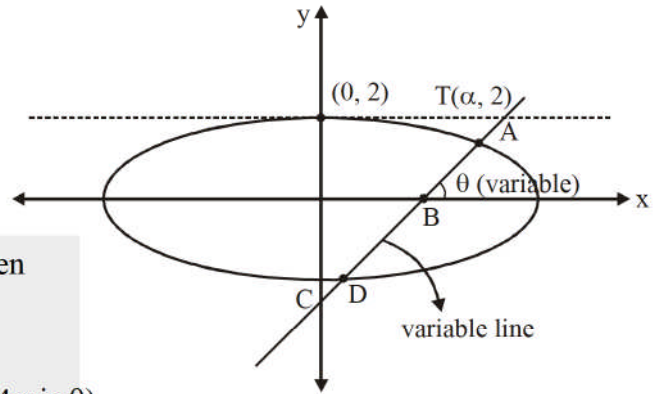
$$\Rightarrow \frac{4\alpha^2}{4\cos^2 \theta + 9\sin^2 \theta} = \frac{2\alpha}{\sin \theta \cdot \cos \theta}$$

But $\alpha \neq 0$ (Think)

$$\text{So, } \alpha = \frac{4 \cot \theta + 9 \tan \theta}{2} \quad (\text{using A.M.} \geq \text{G.M.})$$

$$\therefore |\alpha| \geq \sqrt{4 \times 9} = 6$$

$$\therefore \alpha \in (-\infty, -6] \cup [6, \infty) \quad [\text{Note : } \tan \theta \text{ and } \cot \theta \text{ must have same sign.}]$$



Q.16 (A) $\sqrt{\frac{5}{3}}$ (C) $\sqrt{\frac{5}{2}}$

Sol. Given that $\frac{dx}{dy} = \frac{3y}{2x} \Rightarrow \int 2x dx = \int 3y dy$

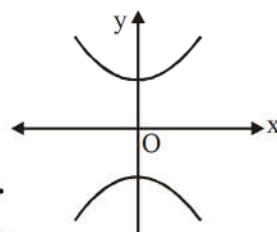
$$\therefore x^2 = \frac{3y^2}{2} + c \quad (\text{But } c \neq 0)$$

$$\Rightarrow \frac{x^2}{c} - \frac{y^2}{2c/3} = 1$$

Now, we shall consider two cases :

Case-I : When $c > 0$

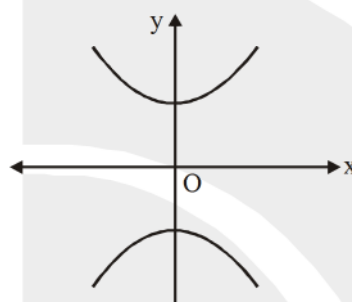
$$\text{Now, } e_h^2 = 1 + \frac{\frac{2c}{3}}{c} = 1 + \frac{2}{3} = \frac{5}{3} \Rightarrow e_h = \sqrt{\frac{5}{3}} \text{ Ans.}$$



Case-II : When $c < 0$:

$$\text{Now, } e_c^2 = 1 + \frac{\frac{c}{2c}}{\frac{3}{2}} = 1 + \frac{3}{2} = \frac{5}{2}$$

$$\therefore e_c = \sqrt{\frac{5}{2}} \text{ Ans.}$$



Q.17 (A) equation of auxiliary circle is $x^2 + y^2 = 2$

(B) eccentricity equals $\sqrt{2}(\sqrt{2}-1)$

Sol. The given ellipse is

$$e^2 = 1 - \frac{2}{4} = \frac{1}{2}$$

$$\frac{x^2}{4} + \frac{y^2}{2} = 1$$

$$\therefore e = \frac{1}{\sqrt{2}}$$

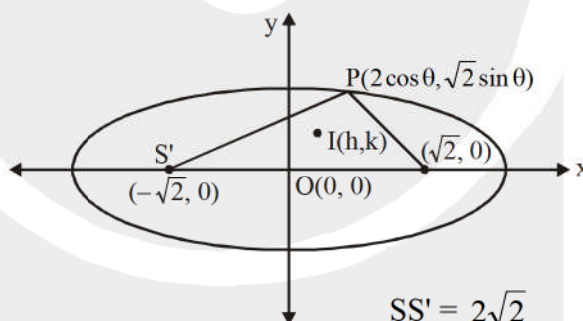
$$S'P = a + e a \cos \theta$$

$$SP = a - e a \cos \theta$$

Now,

$$h = \frac{(2\sqrt{2})(2 \cos \theta) + \sqrt{2}(2 + \sqrt{2} \cos \theta) - \sqrt{2}(2 - \sqrt{2} \cos \theta)}{4 + 2\sqrt{2}}$$

$$\therefore h = \frac{4\sqrt{2} \cos \theta + 2 \cos \theta + 2 \cos \theta}{4 + 2\sqrt{2}} \Rightarrow h = \frac{4(\sqrt{2}+1)\cos \theta}{2(2+\sqrt{2})} \Rightarrow h = \frac{4(\sqrt{2}+1)}{2\sqrt{2}(\sqrt{2}+1)} \cos \theta$$



$$SS' = 2\sqrt{2}$$

Also,

$$PS' + PS = 4$$

$$\therefore h = \sqrt{2} \cos \theta \Rightarrow (\cos \theta) = \frac{h}{\sqrt{2}} \quad \dots (1)$$

$$\text{Similarly, } K = \frac{(2\sqrt{2})(\sqrt{2} \sin \theta) + 0 + 0}{4 + 2\sqrt{2}}$$

$$\Rightarrow K = \frac{4 \sin \theta}{2(2 + \sqrt{2})} \Rightarrow \sin \theta = \frac{(2 + \sqrt{2})K}{2} \quad \dots (2)$$

$$\text{As, } \cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \frac{h^2}{2} + \frac{(2 + \sqrt{2})^2}{4} K^2 = 1$$

$\therefore$ Locus of incentre I(h, k) of $\Delta P'S$ is

$$\frac{x^2}{2} + \frac{y^2}{\frac{4}{(2 + \sqrt{2})^2}} = 1 \quad \text{or} \quad \frac{x^2}{(\sqrt{2})^2} + \frac{y^2}{\left(\frac{2}{2 + \sqrt{2}}\right)^2} = 1$$

The equation of (horizontal ellipse) auxiliary circle is $x^2 + y^2 = 2$

$$\text{Also, } e_1^2 = 1 - \frac{4}{(2 + \sqrt{2})^2 \cdot 2} = 1 - \frac{2}{(2 + \sqrt{2})^2} = \frac{(2 + \sqrt{2})^2 - (\sqrt{2})^2}{(2 + \sqrt{2})^2}$$

$$\therefore e_1^2 = \frac{2(2 + 2\sqrt{2})}{(2 + \sqrt{2})^2}$$

$$\therefore e_1^2 = \frac{4(1 + \sqrt{2})}{2(1 + \sqrt{2})^2} = \frac{2}{\sqrt{2} + 1} \Rightarrow e_1^2 = 2(\sqrt{2} - 1)$$

Hence, $e_1 = \sqrt{2(\sqrt{2} - 1)}$ **Ans.**

Q.18 8

Sol. Upper end of latus-rectum of the parabola $y^2 = 12x$ is (3, 6)

Equation of tangent at (3, 6) to the parabola $y^2 = 12x$ is

$$y \cdot 6 = 2 \cdot 3 \cdot (x + 3)$$

$$y = x + 3$$

Let $A(x_1, y_1)$ be a point on the line $x - y + 3 = 0$

COC of hyperbola is

$$\frac{xx_1}{4} - \frac{yy_1}{1} = 1 \quad \dots (i)$$

Equation of chord whose middle point is (h, k)

$$\frac{hx}{4} - \frac{ky}{1} = \frac{h^2}{4} - \frac{k^2}{1} \quad \dots\dots(ii)$$

From (i) and (ii)

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{1}{\frac{h^2}{4} - \frac{k^2}{1}}$$

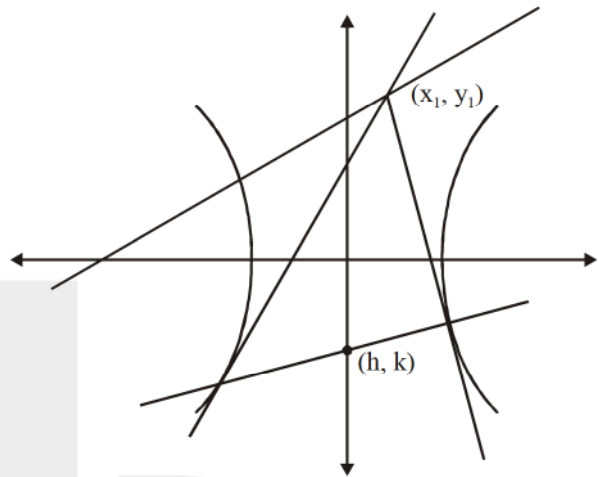
$$\Rightarrow x_1 = \frac{h}{\frac{h^2}{4} - \frac{k^2}{1}}, y_1 = \frac{k}{\frac{h^2}{4} - \frac{k^2}{1}}$$

$$\therefore x_1 - y_1 + 3 = 0$$

$$h - k + 3 \left(\frac{h^2}{4} - \frac{k^2}{1} \right) = 0$$

$$3x^2 - 12y^2 + 4x - 4y = 0$$

$$\therefore a + b = 12 - 4 = 8 \text{ Ans.}$$



Q.19 9

Sol. Let P(h, k), then equation of chord of contact is $\frac{hx}{2} - ky = 1 \quad \dots\dots(i)$

and equation of normal to hyperbola is $\sqrt{2} x \cos \phi + y \cot \phi = 3 \quad \dots\dots(ii)$

Compare (i) and (ii), we get

$\cot \phi = -3k$, $\sec \phi = \frac{2\sqrt{2}}{3h}$, Now eliminate ϕ

we get $\frac{8}{h^2} - \frac{1}{k^2} = 9$

$$\therefore \lambda = 9. \text{ Ans.}$$